# Linear Algebra 2026 Homework 5

Due: October 30, 2026

*Problems should be solved in English and calculation steps are required. For all T/F questions, you don't need to justify your answer.*

*All notations can be found in the book "Linear Algebra — the fourth edition" by Stephen H. Friedberg, Arnold J. Insel, Lawrence E. Spence. If you cannot locate them, feel free to contact the TAs.*

**Problem 1 [10 pts].** Let $\sim$ mean "is isomorphic to." Prove that $\sim$ is an equivalence relation on the class of vector spaces over $F$.

**Problem 2 [10 pts].** Let $\mathsf{V}$ and $\mathsf{W}$ be finite-dimensional vector spaces, and let $\mathsf{T} : \mathsf{V} \to \mathsf{W}$ be a linear transformation. Suppose that $\beta$ is a basis for $\mathsf{V}$. Prove that $\mathsf{T}$ is an isomorphism if and only if $\mathsf{T}(\beta)$ is a basis for $\mathsf{W}$.

**Problem 3 [10 pts].** Let $\mathsf{V}$ and $\mathsf{W}$ be finite-dimensional vector spaces and $\mathsf{T} : \mathsf{V} \to \mathsf{W}$ be an isomorphism. Let $\mathsf{V}_0$ be a subspace of $\mathsf{V}$.

(a) Prove that $\mathsf{T}(\mathsf{V}_0)$ is a subspace of $\mathsf{W}$.

(b) Prove that $\dim(\mathsf{V}_0) = \dim(\mathsf{T}(\mathsf{V}_0))$.

**Problem 4 [5 pts].** Label the following statements as true or false.

(a) Suppose that $\beta = \{x_1, x_2, \ldots, x_n\}$ and $\beta' = \{x'_1, x'_2, \ldots, x'_n\}$ are ordered bases for a vector space and $Q$ is the change of coordinate matrix that changes $\beta'$-coordinates into $\beta$-coordinates. Then the $j$th column of $Q$ is $[x_j]_{\beta'}$.

(b) Every change of coordinate matrix is invertible.

(c) Let $\mathsf{T}$ be a linear operator on a finite-dimensional vector space $\mathsf{V}$, let $\beta$ and $\beta'$ be ordered bases for $\mathsf{V}$, and let $Q$ be the change of coordinate matrix that changes $\beta'$-coordinates into $\beta$-coordinates. Then $[\mathsf{T}]_\beta = Q[\mathsf{T}]_{\beta'}Q^{-1}$.

(d) The matrices $A, B \in \mathsf{M}_{n\times n}(F)$ are called similar if $B = Q^tAQ$ for some $Q \in \mathsf{M}_{n\times n}(F)$.

(e) Let $\mathsf{T}$ be a linear operator on a finite-dimensional vector space $\mathsf{V}$. Then for any ordered bases $\beta$ and $\gamma$ for $\mathsf{V}$, $[\mathsf{T}]_\beta$ is similar to $[\mathsf{T}]_\gamma$.

**Problem 5 [10 pts].** For each of the following pairs of ordered bases $\beta$ and $\beta'$ for $\mathsf{R}^2$, find the change of coordinate matrix that changes $\beta'$-coordinates into $\beta$-coordinates.

(a) $\beta = \{e_1, e_2\}$ and $\beta' = \{(a_1, a_2), (b_1, b_2)\}$

(b) $\beta = \{(-1, 3), (2, -1)\}$ and $\beta' = \{(0, 10), (5, 0)\}$

**Problem 6 [5 pts].** For the following pairs of ordered bases $\beta$ and $\beta'$ for $\mathsf{P}_2(R)$, find the change of coordinate matrix that changes $\beta'$-coordinates into $\beta$-coordinates. $\beta = \{x^2, x, 1\}$ and $\beta' = \{a_2x^2 + a_1x + a_0, b_2x^2 + b_1x + b_0, c_2x^2 + c_1x + c_0\}$

**Problem 7 [10 pts].** Let $\mathsf{T}$ be the linear operator on $\mathsf{R}^2$ defined by

$$\mathsf{T}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2a+b \\ a-3b \end{pmatrix},$$

let $\beta$ be the standard ordered basis for $\mathsf{R}^2$, and let

$$\beta' = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}.$$

Use Theorem 2.23 and the fact that

$$\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$$

to find $[\mathsf{T}]_{\beta'}$.

**Problem 8 [10 pts].** For each matrix $A$ and ordered basis $\beta$, find $[\mathsf{L}_A]_\beta$. Also, find an invertible matrix $Q$ such that $[\mathsf{L}_A]_\beta = Q^{-1}AQ$.

(a) $A = \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix}$ and $\beta = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$

**Problem 9 [15 pts].** Prove the following generalization of Theorem 2.23 (in "Linear Algebra fourth edition"). Let $\mathsf{T} : \mathsf{V} \to \mathsf{W}$ be a linear transformation from a finite-dimensional vector space $\mathsf{V}$ to a finite-dimensional vector space $\mathsf{W}$. Let $\beta$ and $\beta'$ be ordered bases for $\mathsf{V}$, and let $\gamma$ and $\gamma'$ be ordered bases for $\mathsf{W}$. Then $[\mathsf{T}]_{\beta'}^{\gamma'} = P^{-1}[\mathsf{T}]_\beta^\gamma Q$, where $Q$ is the matrix that changes $\beta'$-coordinates into $\beta$-coordinates and $P$ is the matrix that changes $\gamma'$-coordinates into $\gamma$-coordinates.

**Problem 10 [10 pts].** Prove that if $A$ and $B$ are similar $n \times n$ matrices, then $\operatorname{tr}(A) = \operatorname{tr}(B)$. *Hint:* Use Exercise 13 of Section 2.3 (in "Linear Algebra fourth edition").

**Problem 11 [10 pts].** Let $\mathsf{V}$ be a finite-dimensional vector space over a field $F$, and let $\beta = \{x_1, x_2, \ldots, x_n\}$ be an ordered basis for $\mathsf{V}$. Let $Q$ be an $n \times n$ invertible matrix with entries from $F$. Define

$$x'_j = \sum_{i=1}^{n} Q_{ij} x_i \quad \text{for } 1 \le j \le n,$$

and set $\beta' = \{x'_1, x'_2, \ldots, x'_n\}$. Prove that $\beta'$ is a basis for $\mathsf{V}$ and hence that $Q$ is the change of coordinate matrix changing $\beta'$-coordinates into $\beta$-coordinates.

**Problem 12 [10 pts].** Prove the converse of Problem 9: If $A$ and $B$ are each $m \times n$ matrices with entries from a field $F$, and if there exist invertible $m \times m$ and $n \times n$ matrices $P$ and $Q$, respectively, such that $B = P^{-1}AQ$, then there exist an $n$-dimensional vector space $\mathsf{V}$ and an $m$-dimensional vector space $\mathsf{W}$ (both over $F$), ordered bases $\beta$ and $\beta'$ for $\mathsf{V}$ and $\gamma$ and $\gamma'$ for $\mathsf{W}$, and a linear transformation $\mathsf{T} : \mathsf{V} \to \mathsf{W}$ such that

$$A = [\mathsf{T}]_\beta^\gamma \quad \text{and} \quad B = [\mathsf{T}]_{\beta'}^{\gamma'}.$$

*Hints:* Let $\mathsf{V} = \mathsf{F}^n$, $\mathsf{W} = \mathsf{F}^m$, $\mathsf{T} = \mathsf{L}_A$, and $\beta$ and $\gamma$ be the standard ordered bases for $\mathsf{F}^n$ and $\mathsf{F}^m$, respectively. Now apply the results of Problem 11 to obtain ordered bases $\beta'$ and $\gamma'$ from $\beta$ and $\gamma$ via $Q$ and $P$, respectively.

**Problem 13 [10 pts].** Let $\mathsf{V} = \mathsf{P}_1(R)$, and, for $p(x) \in \mathsf{V}$, define $\mathsf{f}_1, \mathsf{f}_2 \in \mathsf{V}^*$ by

$$\mathsf{f}_1(p(x)) = \int_0^1 p(t)\,dt \quad \text{and} \quad \mathsf{f}_2(p(x)) = \int_0^2 p(t)\,dt.$$

Prove that $\{\mathsf{f}_1, \mathsf{f}_2\}$ is a basis for $\mathsf{V}^*$, and find a basis for $\mathsf{V}$ for which it is the dual basis.

**Problem 14 [10 pts].** Let $\mathsf{V} = \mathsf{P}_n(F)$, and let $c_0, c_1, \ldots, c_n$ be distinct scalars in $F$. For $0 \le i \le n$, define $\mathsf{f}_i \in \mathsf{V}^*$ by $\mathsf{f}_i(p(x)) = p(c_i)$. Prove that $\{\mathsf{f}_0, \mathsf{f}_1, \ldots, \mathsf{f}_n\}$ is a basis for $\mathsf{V}^*$. *Hint:* Apply any linear combination of this set that equals the zero transformation to $p(x) = (x - c_1)(x - c_2)\cdots(x - c_n)$, and deduce that the first coefficient is zero.

**Definition 1. Exactness.** Let $A \xrightarrow{S} B \xrightarrow{T} C$ be linear maps between vector spaces. We say that the composition is *exact* if $\ker T = \operatorname{Im} S$. Exact sequences are a basic tool of homological algebra, which has application in many math subjects, including topology, category theory, algebraic geometry, group theory, and number theory.

**Problem 15 [15 pts].** Based on the definition of exactness, solve the following problems:

(a) Suppose $A = 0$ is the zero vector space and $S : A \to B$ is the zero map. Translate what it means for the composition $0 \to B \to C$ to be exact into more familiar terminology.

(b) Suppose instead that $C = 0$ and $T : B \to C$ is the zero map. Translate what it means for the composition $A \to B \to 0$ to be exact into more familiar terminology.

(c) Observe that if $A \xrightarrow{S} B \xrightarrow{T} C$ is exact, then $TS = 0$. Give an example of linear maps $S : \mathbb{R} \to \mathbb{R}^2$ and $T : \mathbb{R}^2 \to \mathbb{R}$ such that $TS = 0$ but the composition $\mathbb{R} \xrightarrow{S} \mathbb{R}^2 \xrightarrow{T} \mathbb{R}$ is not exact.

**Definition 2. Exact sequences.** Let's generalize the notion of exactness. Given a sequence

$$V_n \xrightarrow{T_n} V_{n-1} \xrightarrow{T_{n-1}} \cdots \xrightarrow{T_2} V_1 \xrightarrow{T_1} V_0$$

of vector spaces $V_0, \ldots, V_n$ with linear maps $T_i : V_i \to V_{i-1}$, we say this sequence is *exact* if $\ker T_i = \operatorname{Im} T_{i+1}$ for every $i$.

**Problem 16 [10 pts].** Let $V$ be a vector space with subspace $W \subset V$. Consider the sequence

$$0 \to W \xrightarrow{S} V \xrightarrow{T} V/W \to 0,$$

where $S$ is the inclusion map $S(x) = x$, $T$ is the quotient map $T(v) = v + W$, and the other maps are zero. Show that this sequence is exact.